

DAY 5 TASKS REPORT

1. User Retention Dashboard

The retention dashboard was created to monitor how effectively users return to the application over time. The dashboard helps in identifying user loyalty, churn behavior, and overall platform engagement.

Metric	Purpose	Example
DAU	Daily active users count	5,000 users/day
WAU	Weekly active users count	18,000 users/week
MAU	Monthly active users count	60,000 users/month
Retention Rate	Returning users percentage	75%
Churn Rate	Users leaving the platform	25%
Session Duration	Average time spent	12 minutes

Retention Rate Formula:

$$\text{Retention Rate} = (\text{Returning Users} / \text{Total Users}) \times 100$$

Example Calculation:

Returning Users = 750

Total Users = 1000

Retention Rate = 75%

2. User Engagement Pattern Analysis

User engagement analysis was performed to understand how users interact with the application and which features receive the most attention.

Metric	Analysis
Average Session Time	Users spend around 12 minutes per session
Pages Per Session	Users visit an average of 5 pages

Bounce Rate	32% of users leave after one page
Feature Usage	Video and dashboard sections are highly used
Click-through Rate	Higher CTR observed on recommendation banners

Observations

- Users engage more with personalized content.
- Peak user activity occurs during evening hours.
- Most users abandon the signup process during verification.
- Push notifications increase returning user count.

3. App Conversion Funnel Analysis

A conversion funnel was studied to track how users move from app visits to successful purchases and repeat usage.

Stage	Users	Conversion %
App Visit	10,000	100%
Sign Up	6,000	60%
Email Verification	4,500	45%
Purchase	2,000	20%
Repeat Usage	1,200	12%

Conversion Rate Formula:

$$\text{Conversion Rate} = (\text{Converted Users} / \text{Total Visitors}) \times 100$$

Key Findings:

- Major drop-off observed during email verification.
- Repeat usage rate is comparatively low.
- Simplifying onboarding can improve conversions.

4. Competitor Analytics Strategy Comparison

Competitor analysis was conducted to study successful engagement and retention techniques used by major platforms.

Competitor	Strategy	Benefit
Netflix	Personalized recommendations	Higher watch time

Spotify	AI-generated playlists	Improved engagement
Amazon	Product recommendations	Increased conversions
Instagram	Push notifications	Better retention

Common strategies identified:

- A/B testing
- Personalized content
- Push notifications
- Gamification
- Recommendation systems

5. Weekly Insights Report

The weekly insights report summarizes overall application performance and provides recommendations for improvement.

Metric	This Week	Last Week	Change
Active Users	15,000	13,500	+11%
Retention Rate	74%	69%	+5%
Conversion Rate	18%	15%	+3%
Average Session Time	12 min	10 min	+2 min

Recommendations

- Improve onboarding experience.
- Reduce signup verification complexity.
- Increase personalized recommendations.
- Optimize notification timing.
- Enhance user interface for better engagement.

Tools and Platforms Used

- Google Sheets – Data cleaning and preprocessing
- Power BI – Dashboard creation and visualization
- Google Analytics – User tracking and engagement analysis

Conclusion

The Day 5 analytics tasks helped in understanding user behavior, retention trends, conversion bottlenecks, and competitor strategies. The generated insights can assist organizations in improving user experience, increasing engagement, and maximizing conversions.